THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS UNIT TEST SET C

Chapters 1–4: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials, The World of Numbers, Exploring Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The x-coordinate of a point is its perpendicular distance from the: 1

(a) line $y = x$
(b) origin
(c) y-axis (measured along the x-axis)
(d) x-axis

2. The midpoint of the segment joining $A(-6, -2)$ and $B(2, 8)$ is: 1

(a) $(-2, -3)$ **(b)** $(-2, 3)$ **(c)** $(-4, 6)$ **(d)** $(4, 5)$

3. The point (a, b) coincides with the point (b, a) if and only if: 1

(a) $a = b$ **(b)** $a = 0$ **(c)** $a = -b$ **(d)** $a + b = 0$

4. A straight line $y = ax$ always passes through: 1

(a) the point $(1, 1)$ only
(b) the origin
(c) no fixed point
(d) the y-axis at $(0, a)$

5. The degree of the polynomial $3x^2 - 5x + 7$ is: 1

(a) 3 **(b)** 2 **(c)** 1 **(d)** 4

6. For the line $y = -x - 2$, the slope and the y-intercept are respectively: 1

(a) 1 and -2
(b) -1 and 2
(c) -1 and -2
(d) -2 and -1

7. The value of $(-8) \times (-7)$ is: 1

(a) -56 **(b)** 58 **(c)** 60 **(d)** 56

8. The decimal expansion of $\frac{9}{14}$ is: 1

(a) non-terminating non-repeating
(b) terminating
(c) non-terminating repeating
(d) cannot be determined

9. Which number is both rational and an integer? 1

(a) $\frac{3}{7}$ **(b)** −7 **(c)** π **(d)** $\sqrt{2}$

10. Expand $(7x + 1)^2$: 1

(a) $49x^2 + 1$ **(b)** $49x^2 + 14x + 1$ **(c)** $49x^2 + 7x + 1$ **(d)** $49x^2 + 14x - 1$

11. Factorise $x^2 + 11x + 30$: 1

(a) $(x - 5)(x + 6)$ **(b)** $(x - 6)(x - 5)$ **(c)** $(x + 5)(x + 6)$ **(d)** $(x - 6)(x + 5)$

12. The value of 29^2 using a suitable identity is: 1

(a) 941 **(b)** 831 **(c)** 841 **(d)** 861

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

13. Find the distance between $A(0,\,5)$ and $B(12,\,0)$. 2

14. Find the values of the linear polynomial $5x - 3$ at $x = 0$ and $x = -1$. 2

15. Express $0.1\overline{3}$ in the form $\frac{p}{q}$. 2

16. Expand $\left(x + \frac{y}{2}\right)^2$ using a suitable identity. 2

17. Factorise $s^2 - 11s + 24$ by splitting the middle term. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Show that the triangle with vertices $A(1,\,1)$, $B(7,\,1)$, $C(4,\,5)$ is isosceles. 3

19. Ruby has 3 times as many ₹2 coins as ₹5 coins. If she has ₹88 in all, how many coins of each type does she have? 3

20. Prove that $\sqrt{3}$ is irrational. 3

21. What is the side of the cube whose volume is $p^3 + 6p^2q + 12pq^2 + 8q^3$? 3

22. A farmer cuts a 300 ft fence into two pieces; the longer is four times the shorter. Find the lengths. 3

OR

A learning platform charges a fixed monthly fee plus a cost per module. For 10 modules the bill is ₹400 and for 14 modules ₹500. Find a and b in $y = ax + b$. 3

23. Find five rational numbers between 3 and 4. 3

OR

Express $0.\overline{45}$ as a fraction in lowest terms and verify by dividing. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) Plot the graphs of $y = 2x - 1$, $y = 2x + 1$ and $y = 2x + 5$ on the same axes by choosing points. 5
(ii) What do you observe about the lines?
(iii) Where does $y = 2x + 5$ cut the y-axis?

OR

Case study: A water tank has height $h(t) = 3 - 0.5t$ m of water after t months of summer. 5
(i) Make a table for $t = 0$ to 6.
(ii) Is it growth or decay? Why?
(iii) When does the tank become empty?
(iv) Draw (describe) the graph and state its slope and intercept.

25. **Case study:** Three consecutive square numbers are $(n-1)^2$, n^2, $(n+1)^2$. 5
(i) Add the smallest and largest and subtract twice the middle one for $n = 6$.
(ii) Prove the result is always 2.
(iii) Does the pattern hold for consecutive even-square numbers $(n-2)^2, n^2, (n+2)^2$? Find (smallest + largest) − 2 × middle.

OR

The sum of three numbers is 10, the sum of their squares is 38 and their product is 25. 5
(i) Find $xy + yz + zx$.
(ii) Hence find the sum of their cubes.

END OF QUESTION PAPER — **Total: 50 Marks**